

Physiology: Ventilation / Perfusion Relationships

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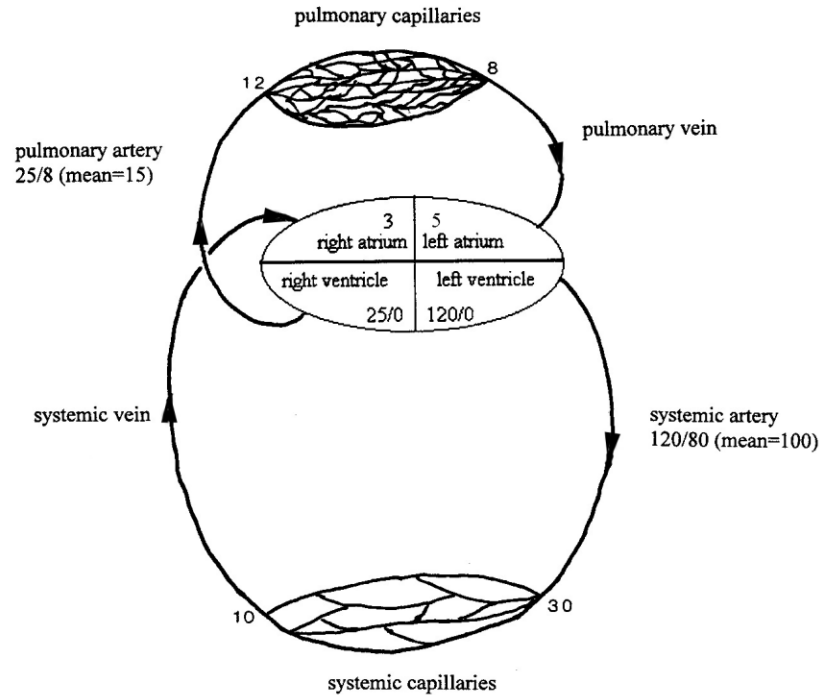


Do.
Make.
Innovate.
Reinvent the Future.

Objectives:

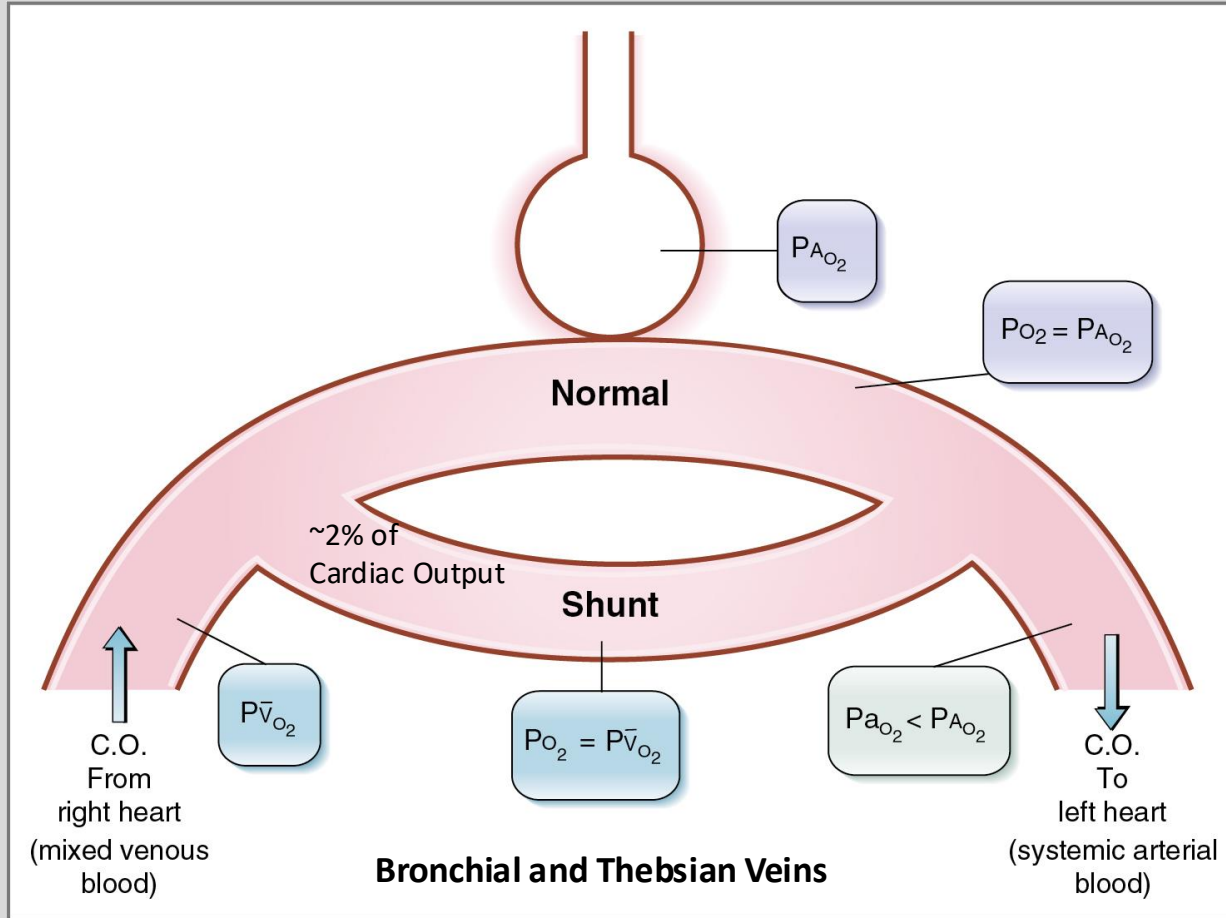
- Compare & contrast physiologic and non-physiologic shunts based on their causes and consequences.
- Explain how the various kinds of shunts respond to supplemental oxygen.
- Compare & contrast focal vs. global hypoxia.
- Compare & contrast the effect of lung volume on alveolar and extra-alveolar vessels.
- Review & recall select pulmonary vasoconstrictors/dilators & airway constrictors.
- Describe the gravitational effects of pulmonary blood flow on the $\dot{V}/\dot{Q}$ at the apex, center and base of the upright lungs and the physiologic challenges that ensue.
- Describe how those differences across the lungs challenge gas exchange.
- Describe how compensatory bronchoconstriction & compensatory hypoxic vasoconstriction can remedy $\dot{V}/\dot{Q}$ mismatches.
- Compare and contrast the major causes of $\dot{V}/\dot{Q}$ defects and the consequences on gas exchange (hypoxemia, hypercarbia).

Pulmonary Circulation - Low Pressure



- Thinner, less muscularized arteries
- Higher vessel compliance
- Less pressure and resistance
- Bronchial & Thebsian veins

Physiologic Right – to – Left Shunts



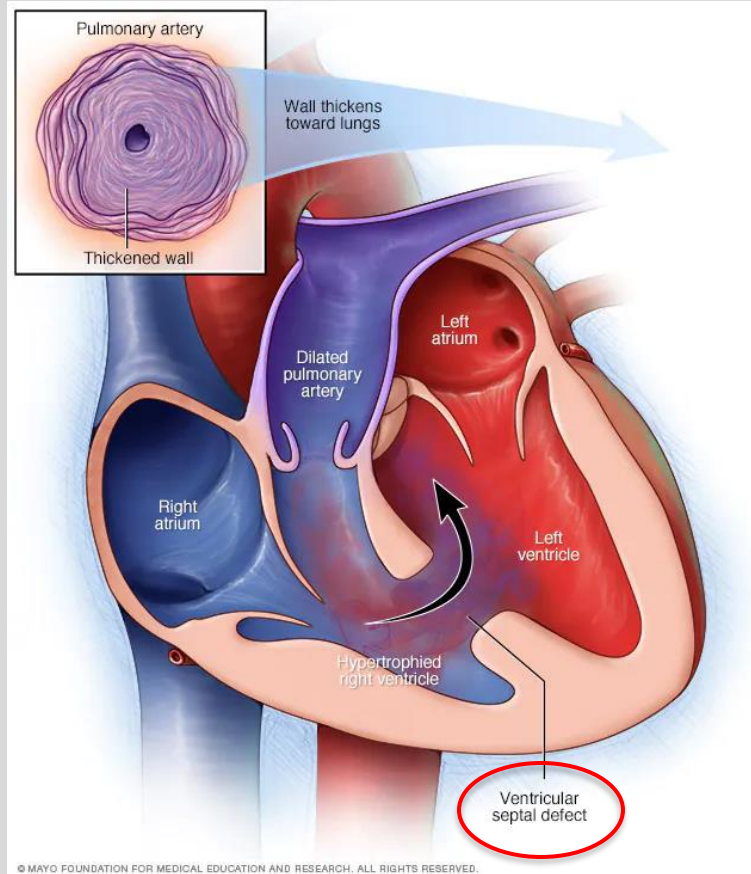
Are there consequences
for Pa_{O_2} ?

How about Pa_{CO_2} ?

Hint: Consider high
sensitivity of CO_2 receptors

Example of a Non-Physiologic Right-to-Left Shunt

Hypoxemia Always Occurs – Cannot be Corrected with Breathing O₂



Eisenmenger Syndrome and others

Shunt fraction equation:

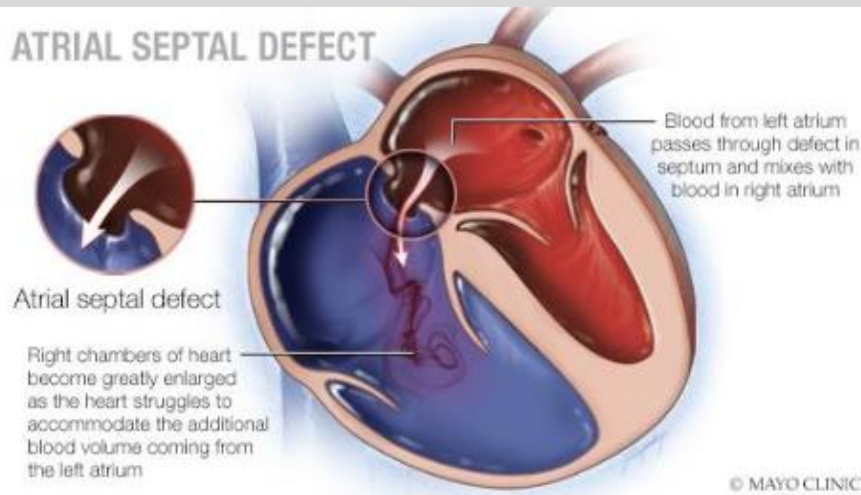
$$\frac{Q_s}{Q_T}$$

QS = Shunted Blood; QT= Cardiac Output

Left-to-Right Shunts

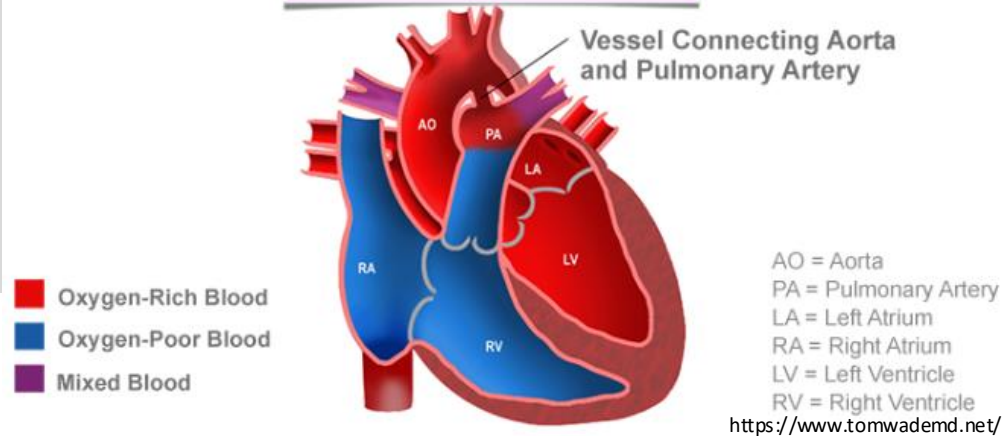
Each ~ 15-20% of Congenital Heart Defects
Do **NOT** cause Hypoxemia

ATRIAL SEPTAL DEFECT



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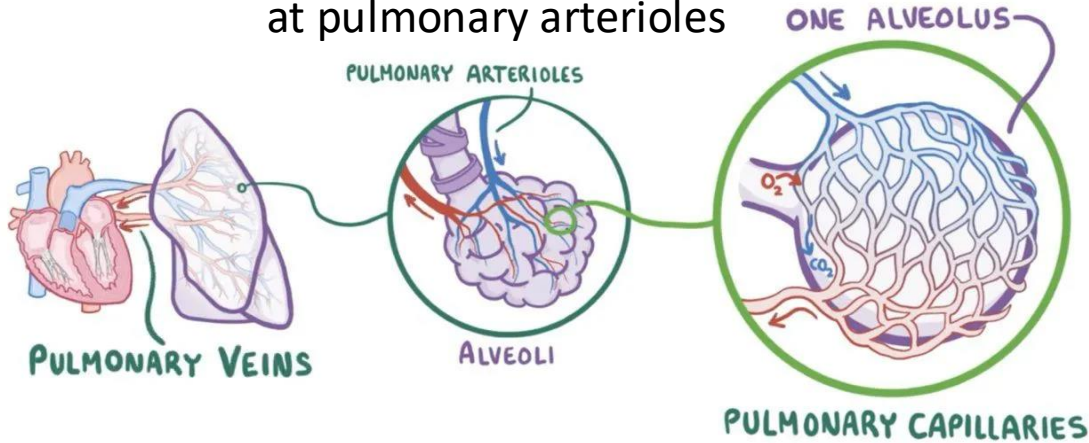
Patent Ductus Arteriosis



Regulation of Pulmonary Blood Flow

Principal Mechanism: Partial Pressure of O₂ in Alveolar Gas

Regulation of resistance
at pulmonary arterioles



https://www.osmosis.org/learn/Regulation_of_pulmonary_blood_flow

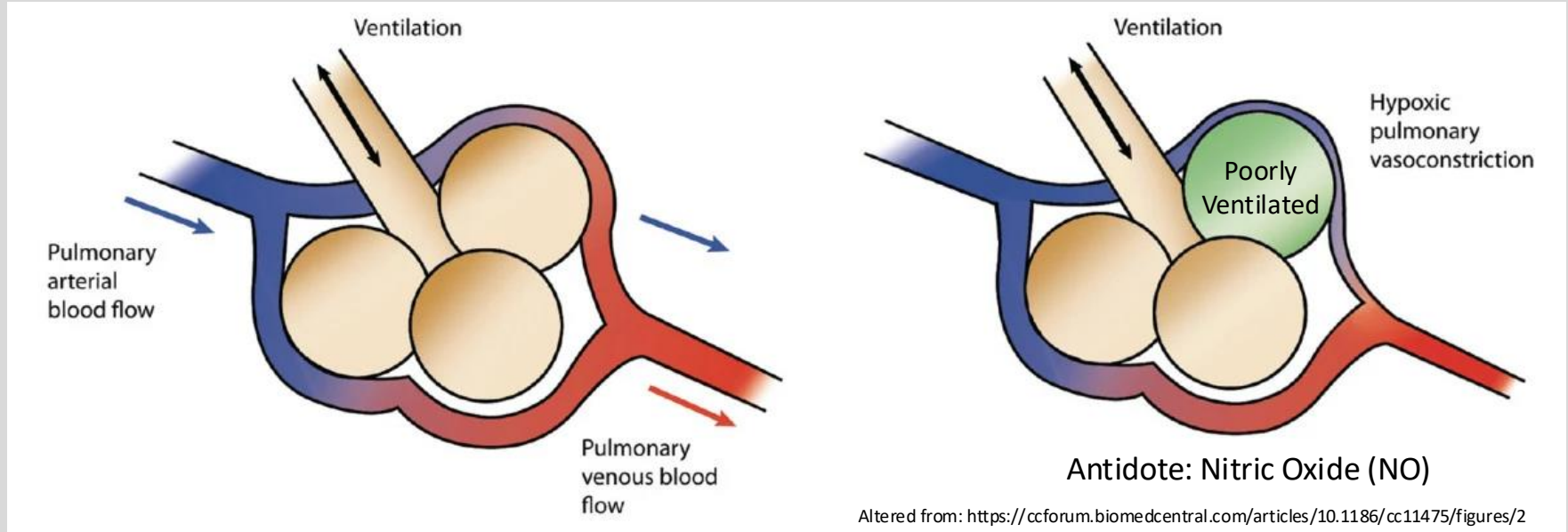
Hypoxic vasoconstriction

- Opposite to most capillary systems
- Prevents perfusion of poorly ventilated areas
- Can be protective (e.g., pneumonia)
- O₂ is lipophilic -> Enters vascular smooth muscle cells
- Depolarization probably involving Ca²⁺ channels

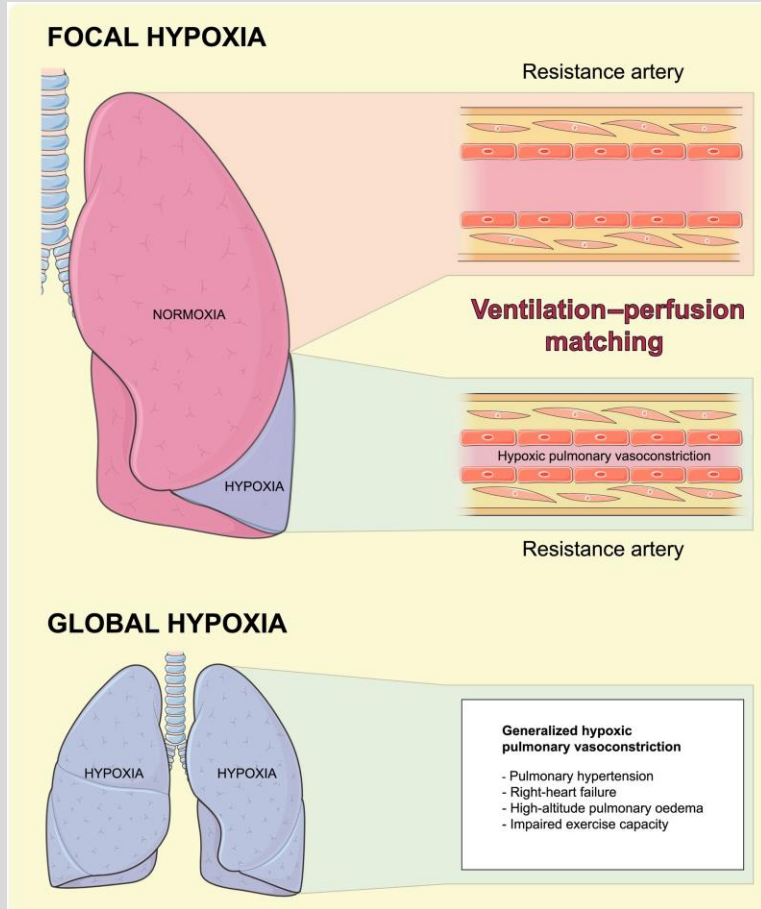
Hypoxic Vasoconstriction – Correcting Mismatches

At PAO_2 below 70mmHg

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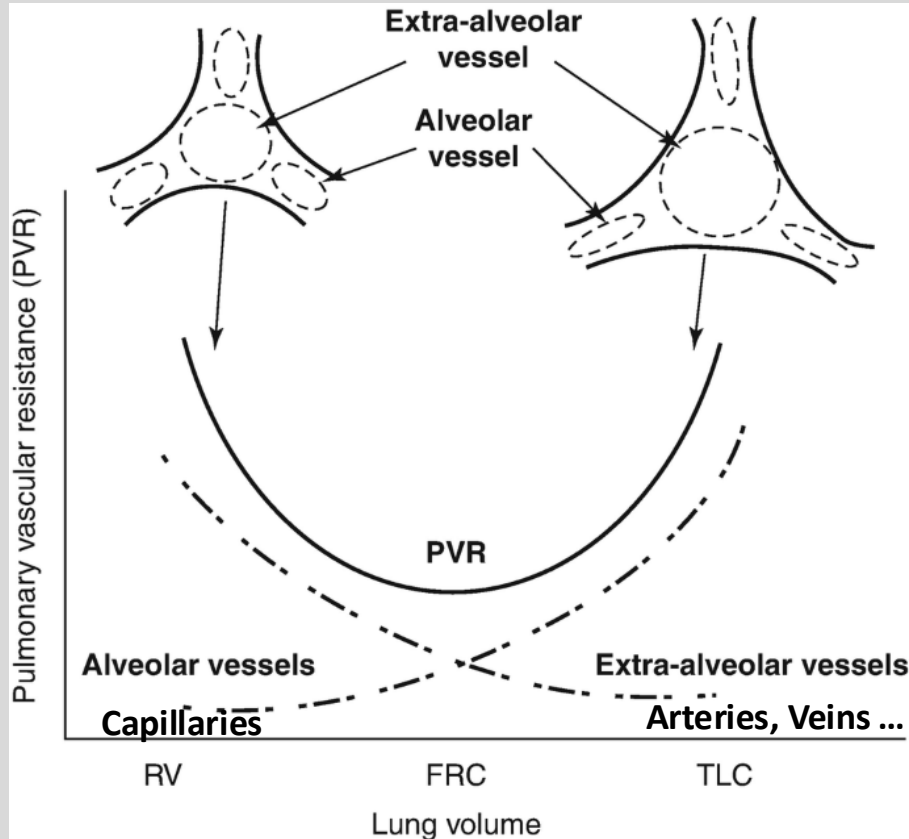
Focal vs Global Hypoxia



No significant increase in in overall vascular resistance

E.g., high altitude, lungs of a fetus
If pathologic, may significantly alter vascular resistance

Lung Volume & Pulmonary Vascular Resistance - Reminder



Lung volume effects **extra-alveolar** and **alveolar** blood vessels in **opposite ways**

Active Regulation of Blood Flow - Select Pulmonary Vasoconstrictors/Dilators & Airway Constrictors – **Review** (Pulmonary Circulation)

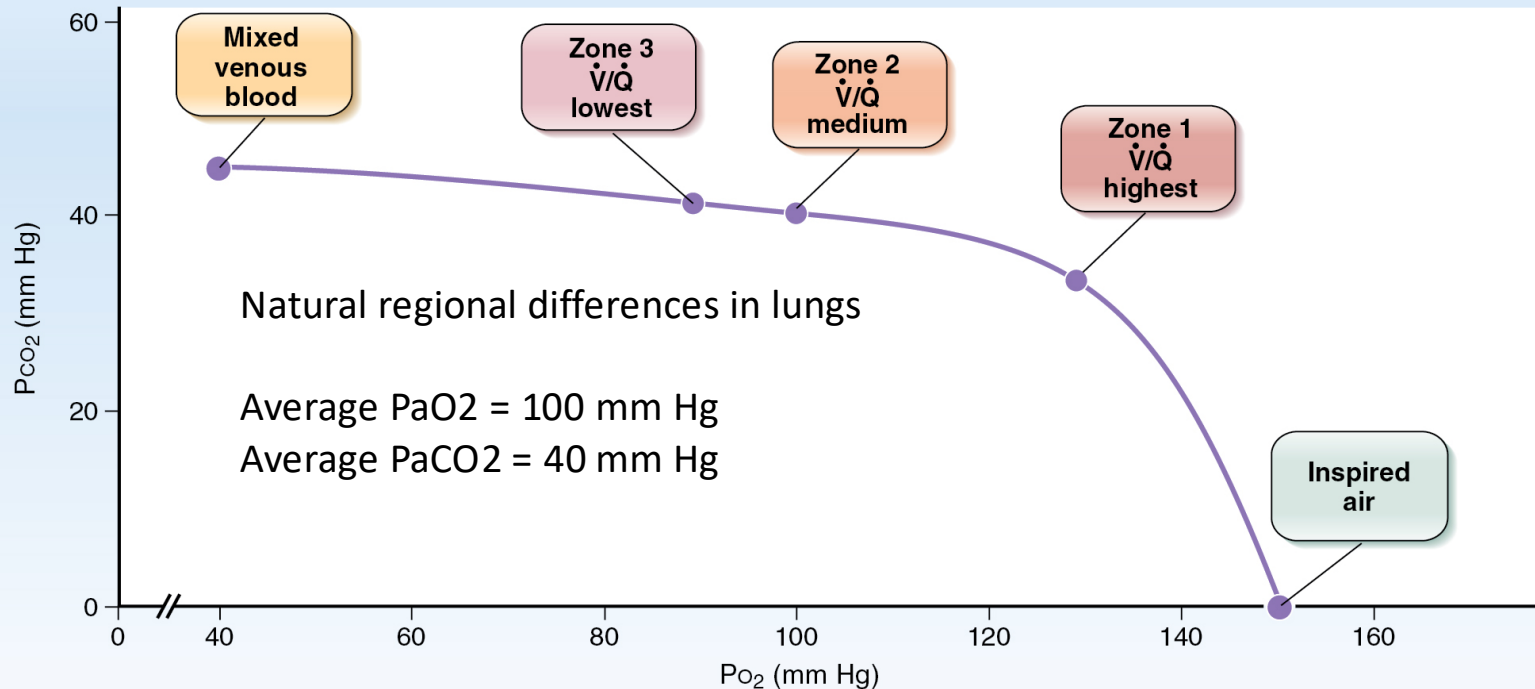
- **Endothelin** - a **potent vasoconstrictor** released from pulmonary endothelial cells in response to shear stress and hypoxia → Major contributor to pulmonary hypertension
- **Thromboxane A2** - a **potent vasoconstrictor** (arachidonic acid derivative) released from white blood cells, macrophages, and platelets
- **Prostacyclin** (prostaglandin I₂, PGI₂; arachidonic acid derivative) - a coagulant & **potent vasodilator** released from pulmonary endothelial cells
- **Nitric oxide (NO)** - a **potent vasodilator** released from capillary endothelial cells in response to acetylcholine, ATP, serotonin, bradykinin, histamine, and substance P
- Leukotrienes (inflammatory mediators) - **potent airway constrictors** released by leukocytes

$\dot{V}/\dot{Q}$ DISTRIBUTION IN THE LUNG

	Blood Flow ($\dot{Q}$)	Alveolar Ventilation ($\dot{V}$)	$\frac{\dot{V}}{\dot{Q}}$	Pa_{O_2}	Pa_{CO_2}
Apex					
Zone 1	Lowest	Lower	Highest (3.0)	Highest (130 mm Hg)	Lower (28 mm Hg)
Zone 2	—	—	—	—	—
Zone 3	Highest	Higher	Lowest (0.6)	Lowest (89 mm Hg)	Higher (42 mm Hg)
Base					

Gravitational Effects

EFFECT OF $\dot{V}/\dot{Q}$ ON GAS EXCHANGE

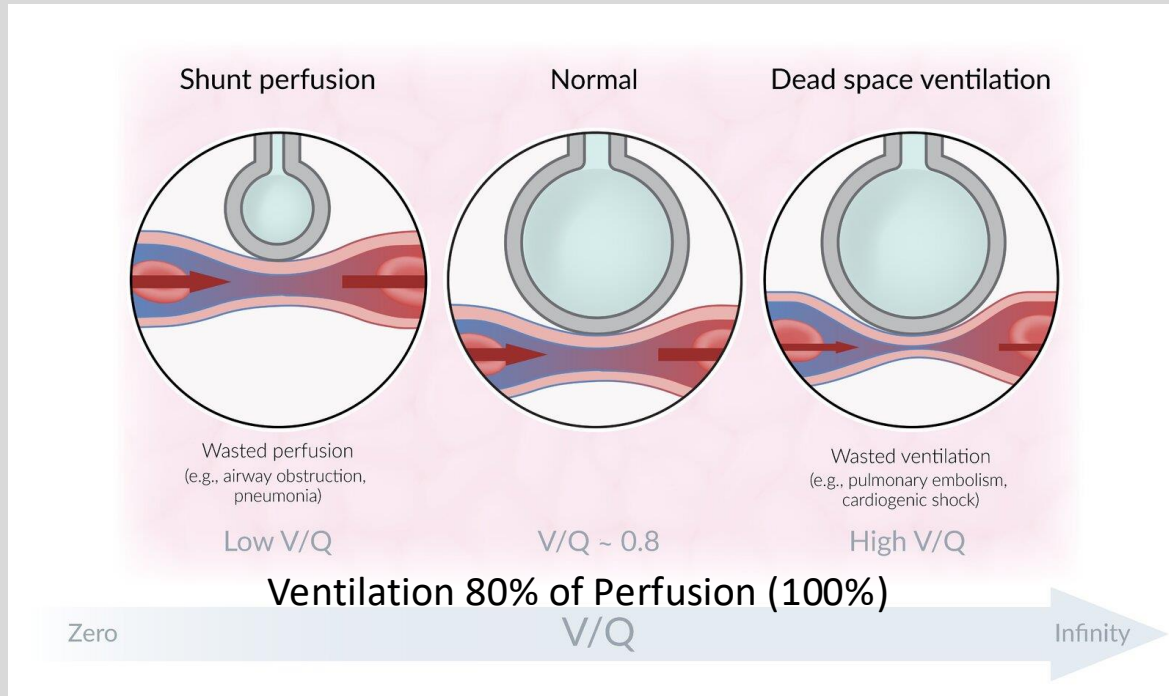


Ventilation/Perfusion Ratios

$\dot{V}/\dot{Q}$ Matching

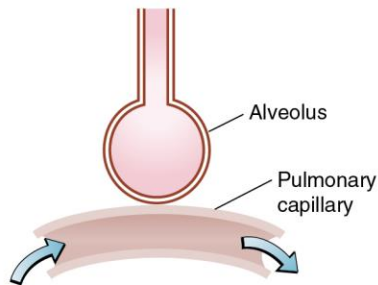
Two Mechanisms:

Compensatory **Bronchoconstriction** & Compensatory **Hypoxic Vasoconstriction**



$\dot{V}/\dot{Q}$ DEFECTS

$\dot{V}/\dot{Q}$ matching

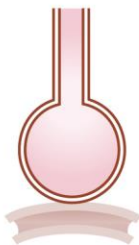


Dead space

$$\dot{V}/\dot{Q} = \infty$$

$$P_{A_{O_2}} = P_{I_{O_2}}$$

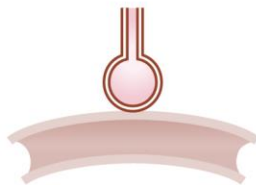
$$P_{A_{CO_2}} = P_{I_{CO_2}}$$



High $\dot{V}/\dot{Q}$

$$\uparrow P_{a_{O_2}}$$

$$\downarrow P_{a_{CO_2}}$$



Low $\dot{V}/\dot{Q}$

$$\downarrow P_{a_{O_2}}$$

$$\uparrow P_{a_{CO_2}}$$



Right-to-left shunt

$$\dot{V}/\dot{Q} = 0$$

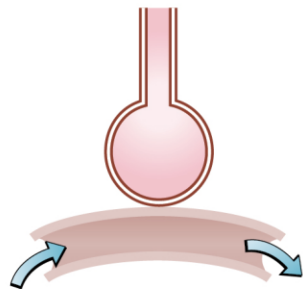
$$P_{a_{O_2}} = P_{\bar{V}_{O_2}}$$

$$P_{a_{CO_2}} = P_{\bar{V}_{CO_2}}$$

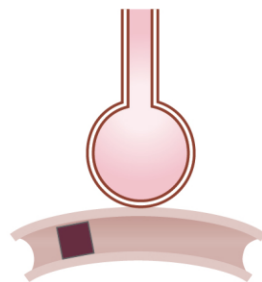
$\dot{V}/\dot{Q}$ defects **always** cause hypoxemia (decreased $P_{a_{O_2}}$) & hypercapnia (increased $P_{a_{CO_2}}$)

$\dot{V}/\dot{Q}$ DEFECTS

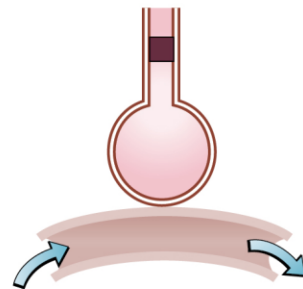
Normal



Pulmonary embolus
(dead space)



Airway obstruction
(shunt)



$\dot{V}/\dot{Q}$	0.8	∞	0
P_{AO_2}	100	150	—
P_{ACO_2}	40	0	—
P_{aO_2}	100	—	40
P_{aCO_2}	40	—	46

A 32-year-old female is admitted to the intensive care unit with acute respiratory distress syndrome (ARDS) secondary to severe atypical pneumonia. Her chest X-ray shows diffuse, bilateral alveolar infiltrates and extensive fluid-filled, collapsed alveoli (atelectasis), creating a massive physiological right-to-left shunt. The resident on duty places the patient on 100% supplemental oxygen via a non-rebreather mask, but her arterial oxygen saturation (S_aO_2) remains critically low at 82%. Which of the following best explains why her hypoxemia is highly resistant to supplemental oxygen therapy?

- A. The high concentration of oxygen causes widespread pulmonary vasodilation, which increases the shunt fraction.
- B. Supplemental oxygen decreases the affinity of hemoglobin for any remaining oxygen in the systemic circulation.
- C. The inspired oxygen cannot reach the blood flowing through the collapsed, unventilated alveoli.
- D. Oxygen gas is highly insoluble in blood plasma, preventing any increase in dissolved oxygen content.
- E. The elevated oxygen levels inhibit the enzyme carbonic anhydrase, blocking normal gas transit.

Resources

Updated resources can be found in the Class Preparation Guide (CPG) that goes with this lecture.

Lecture Feedback Form

<https://www.surveymonkey.com/r/NYITCOMLectureFeedback>